

Stack Overflow Developer Survey — Data Cleaning & Power BI Modelling Cheat Sheet

1. Data Cleaning & Transformation

1. Initial Data Assessment

- Started with 49,191 survey responses and approximately 170 columns.
- Selected approximately 38 relevant columns for the planned analysis.
- Reviewed data types, missing values, duplicate ResponsesIds, extreme values, and multi-valued fields.

2. Data Type Standardization

Using Power Query:

- Converted relevant numerical fields from Text → Number.

3. Duplicate Validation

Checked ResponsesId using:

- Excel → Conditional Formatting → Highlight Cells Rules → Duplicate Values
- Power Query → Group By → ResponsesId → Count Rows

Result: No duplicate ResponsesId records were identified.

4. Removal of Essentially Incomplete Responses

- Initial responses: 49,191
- Removed 852 responses containing virtually no useful analytical information.
- Final responses: 48,339.
- Rows were not removed simply because an individual field was blank.

5. Work Experience & Years of Coding

Created cleaned analytical columns:

- WorkExperience_Clean
- YearsCode_Clean

Process:

- Extreme/impossible values were identified and converted to blank/null.
- Valid values were retained.
- After outlier cleaning, remaining missing values were imputed using the median.
- Age-consistency checks were performed using ExperienceConsistency and CodeConsistency.

6. Age Consistency Checks

Created:

- ExperienceConsistency
- CodeConsistency

Possible results:

- OK
- Potentially Inconsistent
- No Check

These checks were used to identify potentially suspicious values rather than automatically declaring them incorrect.

2. Compensation Cleaning & Missing Values

7. Compensation Outlier Cleaning

Compensation was cleaned before salary analysis.

Condition	Treatment
Missing compensation	Blank/null
$\leq \$1,000$	Removed from salary analysis, except qualifying student cases
Student + compensation $> \$500,000$	Blank/null
Not employed + compensation $> \$250,000$	Blank/null
$> \$5,000,000$	Blank/null
Remaining valid values	Retained

8. Rows Without Compensation

After compensation cleaning, respondents with no usable compensation were removed from the salary-analysis dataset.

3. Multi-Valued Data & Data Modelling

9. Handling Multi-Valued Survey Questions

Several survey fields contained multiple selections in a single cell, separated by semicolons.

The data was transformed in Power Query using:

- Split Column → By Delimiter → Semicolon (;) → Split into Rows

This converted each individual selection into a separate row while retaining the respondent identifier.

10. Factless Tables

The multi-valued data was separated into the required factless tables:

- Factless_Language
- Factless_Database
- Factless_AIModel
- Factless_Platform
- Factless_AIAgent

Factless_Platform and Factless_AIAgent were normalized the same way as the other multi-valued fields (Split Column → By Delimiter → Split into Rows), covering the platforms and AI-agent usage questions respectively.

11. Not Available Handling

"Not Available" values were reviewed after the multi-valued transformation.

Where "Not Available" did not represent a genuine analytical entity, it was removed using:

- Power Query → Filter Rows → Text/Value Filter

The respondent itself was not deleted; only the non-meaningful relationship was removed from the relevant factless table.

12. Dim_Respondent

Created:

- Dim_Respondent

Containing:

- RespondentKey — newly created unique key
- Responseld — original survey identifier

RespondentKey was used as the relationship key.

13. Why Dim_Respondent Was Created

- A respondent can have multiple records in a factless table.
- Connecting these tables directly to Fact_Survey could duplicate the respondent or create inappropriate many-to-many relationships.
- Dim_Respondent therefore acts as the common respondent-level connection.

14. Final Relationship Structure

- Dim_Respondent connects to the relevant fact and factless tables using RespondentKey.
- The factless tables connect to their corresponding dimensions using their entity keys.
- Factless_AIAgent connects only at the respondent level, since AI-agent usage does not require a separate dimension table.

Intended relationship pattern:

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Dim_Respondent ---< Fact_Survey
Dim_Respondent ---< Factless_Language >--- Dim_Language
Dim_Respondent ---< Factless_Database >--- Dim_Database
Dim_Respondent ---< Factless_AIModel >--- Dim_AIModel
Dim_Respondent ---< Factless_Platform >--- Dim_Platform
Dim_Respondent ---< Factless_AIAgent

```

15. Main Power Query Features Used

Power Query Feature	Purpose
Change Type	Converted fields to appropriate data types
Conditional/Custom Column	Created cleaned and derived fields
Split Column → By Delimiter	Separated multi-valued responses
Split into Rows	Converted multiple selections into individual records
Filter Rows	Removed invalid/unwanted values
Group By	Checked duplicate ResponsesIds
Reference	Created fact and dimension queries from the main Data query
Remove Columns	Removed unnecessary fields
Rename Columns	Standardized column names
Add Index Column	Created unique respondent keys where required